

Find the gradients of the following functions:

(a) $f(x, y, z) = x^2 + y^3 + z^4$

(b) $f(x, y, z) = x^2 y^3 z^4$

(c) $f(x, y, z) = e^x \sin(y) \ln(z)$

Solution:

The gradient of a function T is defined as

$$\nabla T = \frac{\partial T}{\partial x} \hat{x} + \frac{\partial T}{\partial y} \hat{y} + \frac{\partial T}{\partial z} \hat{z}$$

This results in

(a) $2x\hat{x} + 3y^2\hat{y} + 4z^3\hat{z}$

(b) $2xy^3z^4\hat{x} + 3x^2y^2z^4\hat{y} + 4x^2y^3z^3\hat{z}$

(c) $e^x \left(\sin(y) \ln(z) \hat{x} + \cos(y) \ln(z) \hat{y} + \frac{\sin(y)}{z} \hat{z} \right)$

□
